III. Give short answers to the following questions

27. Write the uses of time series.

28. Explain semi-averages method

29. Write the merits of moving averages.

30. What is cyclical variation?

31. What is seasonal variation?

32. What are medium-term and long-term forecasts?

33. Describe the method of finding seasonal indices.

34. With what characteristic component of a time series should each of the following be associated.
    (i) An upturn in business activity
    (ii) Fire loss in a factory
    (iii) General increase in the sale of Television sets.

35. The number of units of a product exported during 1990-97 is given below. Draw the trend line using graphical method.

Year	1990	1991	1992	1993	1994	1995	1996	1997
No. of units exported (in '000)	12	13	13	16	19	23	21	23

36. Draw a time series graph relating to the following data and show the trend by free hand method

Year	1996	1997	1998	1999	2000	2001	2002	2003
Production in Million tonnes	40	44	42	48	51	54	50	56

37. Draw a trend line by the method semi averages

Year	1992	1993	1994	1995	1996	1997
Production of steel in Million tonnes	21	23	25	23	26	25

38. Yield of ground nut in Kharif season in India for the years 2003-04 to 2009-10 are given below. Calculate 3-year moving averages.

Year	2003-04	2004-05	2005-06	2006-07	2007-08	2008-09	2009-10
Yield (kg/hectare)	1320	909	1097	689	1386	1063	835

39. What do you understand by seasonal variations?

12th_Statistics_EM_Unit_7.indd 203 3/4/2019 9:39:35 AM